

N.S. LABORATORY SDN. BHD.

CO-MAXAZOL, 80/400 mg/ml, Liquid suspension

Active Ingredients:

Each ml contains

Trimethoprim-----80mg

Sulfamethoxazole-----400mg

Product Description:

White colour cloudy liquid

Dosage Form:

Liquid suspension

Pharmacodynamics:

In vitro trimethoprim is generally bacteriostatic and has a broad spectrum of activity against both gram-positive and gram-negative bacteria. A synergistic and bactericidal effect occurs when trimethoprim is combined with sulfamethoxazole, because trimethoprim and sulfamethoxazole inhibit sequential steps in the synthesis of tetrahydrofolic acid, an essential metabolic cofactor in bacterial synthesis of purine and, subsequently, DNA.

Pharmacokinetics:

Following oral administration both active ingredients are rapidly absorbed from the gut. High trimethoprim concentrations are found in the kidneys, the liver and the lungs. With the exception of the kidneys, sulfamethoxazole concentrations in the tissues are significantly lower than in the plasma. Protein binding for trimethoprim and sulfamethoxazole is not very high. The drugs are primarily excreted through the kidneys (both actively and passively), but elimination also occurs through the faeces. Elimination is relatively fast both in poultry and pigs. Within 48 hours after the last medication trimethoprim, sulfamethoxazole and their metabolites are undetectable in urine and faeces.

Indication:

Pigs: Treatment and metaphylaxis of respiratory infections caused by *Actinobacillus pleuropneumoniae* susceptible to trimethoprim and sulfamethoxazole where the disease has been diagnosed in the herd.

Broilers: Treatment and metaphylaxis of respiratory infections caused by *Escherichia coli* susceptible to trimethoprim and sulfamethoxazole where the disease has been diagnosed in the flock.

Mode of Administration:

Oral administration via drinking water.

Recommended Dosage:

Route of administration: in drinking water use. The product can be added directly to the drinking water to prepare a therapeutic solution at the calculated concentration.

Pigs- 25 mg TMPs per kg bodyweight per day for 3-4 days, corresponding to 1 ml of the veterinary medicinal product per 19.2 kg bodyweight per day. This corresponds to approximately 1 litre of the veterinary medicinal product in 2000 L drinking water. The exact daily amount of the veterinary product can be calculated according to the following formula:

$$\frac{\text{Mean body weight (kg) of pigs to be treated}}{\text{Mean daily water consumption (l) per pig} \times 19.2} = \frac{\text{xx ml veterinary product}}{\text{l drinking water}}$$

Broilers- 33 mg TMPs per kg bodyweight per day for 3-4 days, corresponding to 1 ml of the veterinary medicinal product per 14.56 kg bodyweight per day. This corresponds to approximately 1 litre of the veterinary medicinal product in 3000 L drinking water. The exact daily amount of the veterinary product can be calculated according to the following formula:

$$\frac{\text{Mean body weight (kg) of broilers to be treated}}{\text{Mean daily water consumption (l) per bird} \times 14.56} = \frac{\text{xx ml veterinary product}}{\text{per l drinking water}}$$

The daily amount is to be added to the drinking water such that all medication will be consumed in 24 hours. Medicated drinking water should be freshly prepared every 24 hours. During the treatment period animals should not have access to other water sources than the medicated water. To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing. The uptake of medicated water depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of the veterinary medicinal products has to be adjusted accordingly.

Contraindications:

Do not use in animals suffering from severe liver or kidney disease, oliguria or anuria. Do not use in animals with impaired haematopoietic systems. Do not use in case of known hypersensitivity to sulfonamides or trimethoprim or to any of the excipients.

Special Warnings for each Target Species:

Severely diseased animals can have a decreased appetite and water consumption. If necessary the concentration of the VMP in the drinking water should be adjusted to make sure that the recommended dosage is being consumed. However if the concentration of the product is increased too much, the intake of the medicated

drinking water decreases for palatability reasons. Therefore water intake should be monitored regularly, especially in broilers. In case of insufficient uptake of water, pigs should be treated parenterally.

Special Precautions for Use in Animals:

Official and local antimicrobial policies should be taken into account when the product is used. Due to the likely variability (time, geographical) in susceptibility of bacteria for potentiated sulphonamides, occurrence of resistance of bacteria may differ from country to country and even from farm to farm, and therefore bacteriological sampling and susceptibility testing are recommended. Use of the product should be based on culture and sensitivity of micro-organisms from diseased cases on farm or from recent previous experience on the farm. Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to sulfamethoxazole and trimethoprim and may also decrease the effectiveness of combinations of trimethoprim with other sulphonamides due to the potential for cross resistance.

Special Precautions to be taken by the Person Administering the Medicinal Products to Animals:

This product contains sulfamethoxazole, which can cause allergic reactions in some people. People with known hypersensitivity (allergy) to sulphonamides or trimethoprim should avoid contact with the veterinary medicinal product. During preparation and administration of the medicated drinking water, skin contact with the drug should be avoided. Therefore it is recommended to wear impermeable gloves e.g. rubber or latex when applying the product. Do not handle this product if you know you are allergic to trimethoprim or sulphonamides. Do not smoke, drink or eat when handling the product. If you develop symptoms following exposure to the product such as skin rash, you should seek medical advice and show the present warning to the doctor. Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention. This product can cause eye-irritation. Always wear protective glasses when mixing the product with drinking water. In the event of eye contact, rinse the eye with copious amounts of clean water and if irritation occurs, seek medical attention. In the event of accidental ingestion, seek medical advice. Wash hands and contaminated skin immediately after handling the product.

Interactions with other Medicaments:

Do not combine with other veterinary medicinal products.

Usage during Pregnancy and Lactation:

Laboratory studies in rats conducted with trimethoprim have shown evidence of teratogenicity at higher doses than recommended therapeutic ones. The safety of the product during pregnancy and lactation has not been established in sows. The use of the product is not recommended in sows during pregnancy and lactation.

Adverse Effects:

A diminished water intake in chickens may occur occasionally. Hypersensitivity reactions can occur rarely (more than 1 but less than 10 animals in 10,000 animals treated).

Overdose and Treatment:

In pigs, a 2 ½ fold overdose induces no adverse reactions. In chicken an acute overdose will not occur because the birds will be reluctant to drink the strongly concentrated drinking water. Chronic overdose in chicken will result in a strongly diminished water and feed intake and retarded growth.

Withdrawal Period (Meat and Offal):

Pigs- 5 days; Broilers- 6 days; Eggs- Not for use in birds producing or intended to produce eggs for human consumption.

Storage Condition:

Store in cool, dark place (< 30°C). Do not freeze.

Shelf Life:

36 months

Shelf Life after Opening:

Discard any unused liquid immediately after opening

Shelf Life after Dilution/Reconstitution:

24 hours

Packaging Available:

1L/bottle; 12 bottles/carton; 12kg/carton

Manufacturer and Registration Holder:

N.S. Laboratory Sdn. Bhd.

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26/06/2024